



EDUCATION

Sri Eshwar College of Engineering	B.E. (ECE) CGPA - 7.71 (up to 5 th sem)	2023 - 2027
Lisieux Matriculation Higher Secondary School	HSC 83.33%	2022 - 2023
Whitefield Global School	SSLC 83.8%	2020 - 2021

INTERNSHIP

EMBEDDED SYSTEMS AND IOT – PUMO TECHNOVATION **Jan 2026 – Feb 2026****Timer Based Password Authentication and Motor Control Using PIC Microcontroller****Tools Used: PIC16F877A, servomotor, 4x4 keypad, 16x2 I2C LCD display, LED's**

Developed a secure motor control system using a PIC16F877A microcontroller with a 4×4 keypad for password input, LCD I2C display for user interface, and a servomotor for controlled access based on timer-based authentication.

STM32: HAL, BARE-METAL PROGRAMMING AND RTOS – AXISWIN TECHNOLOGIES **July 2025****Real-Time Temperature Monitoring with STM32****Tools Used: STM32, HAL, FreeRTOS, I2C LCD, Temperature Sensor, STM32CubeIDE, Embedded C**

Built a real-time STM32-based temperature monitoring system using FreeRTOS, processing sensor data and displaying it on an I2C LCD with multitasking support.

EMBEDDED SYSTEMS AND IOT - CSIR-CSIO, CHENNAI**July 2025 – Aug 2025****Abstraction and Detection System in Vehicles****Tools Used: Arduino IDE, ESP32, Ultrasonic Sensors, LED's, Active Buzzer, general purpose PCB board, 2.4-inch SPI 240x320 TFT display**

Developed an ESP32-based real-time obstacle detection system integrating 4 ultrasonic sensors and a 2.4" SPI TFT display with LED/buzzer alerts to indicate distance and safety zones.

EMBEDDED SYSTEMS AND IOT – PUMO TECHNOVATION**Jan 2025 – Feb 2025****Smart Soil Sensor for determining Constructability and Foundation Design (Mini – Project)****Tools Used: Arduino IDE, Arduino UNO, NodeMCU, Soil Moisture Sensor, LED's**

Designed a Smart Soil Sensor using Arduino and NodeMCU to assess soil moisture for construction suitability with real-time LED and remote monitoring.

EMBEDDED IOT AND ROBOTICS – ADVERK TECHNOLOGIES**April 2024 – June 2024****Door Lock Security System (Mini – Project)****Tools Used: Tinkercad, Arduino IDE**

Simulated an automatic door lock system in Tinkercad using Arduino, keypad, servo motor and buzzer for touchless entry with audio feedback.

Attendance Tracker using IoT and RFID card (Major – Project)**Tools Used: Arduino IDE, NodeMCU, RFID Reader, Arduino UNO**

Developed an IoT-based RFID attendance tracker using Arduino, NodeMCU, and RFID reader for real-time cloud-based monitoring.

MERN STACK – RV TECHLEARN**January 2025****To Do List (Mini - Project)****Tools Used: MERN Stack (MongoDB, Express.js, React.js, Node.js)**

Developed a MERN stack To-Do List app with CRUD operations, RESTful APIs and an intuitive UI for efficient task management.

PROJECTS**PEDITRIX: Predictive Industrial Maintenance System****March 2026****Technologies: ESP32, DS18B20, SW-420, 16x2 I2C LCD display, Python, Random Forest, Flutter, MongoDB, REST API**

Built an IoT-based predictive maintenance solution that collects machine sensor data and uses a Random Forest model to predict equipment failures, with real-time monitoring through a mobile application.

FLEETAI: IOT- POWERED SMART FLEET WITH TINYML INTELLIGENCE**Nov 2025****Technologies: ESP32, GPS Module, DHT11, Arduino IDE, TinyML (TensorFlow Lite), Python, Dash, Flutter, REST APIs, OpenStreetMap**

Developed a real-time IoT and TinyML-based fleet tracking system with GPS tracking, sensor monitoring, and intelligent alerts, integrated with a web dashboard and Flutter mobile app for efficient fleet management.

RASPBERRY PI USING STAR DELTA

April 2025

Technologies: Raspberry Pi, Python, GPIO, Relay Modules, Electrical Control Systems

Designed a motor control system using the Star-Delta method with Raspberry Pi, Python and relays to automate safe startup and reduce inrush current.

ATTENDANCE TRACKER USING RFID

February 2025

Technologies: RFID Reader, RFID Tags, Arduino UNO, Arduino IDE, LCD Display, Google Sheets, IoT (Google Apps Script)

Developed an automated RFID attendance system using Arduino UNO and IoT integration to log data on an LCD and update Google Sheets in real time.

WEATHER APP

April 2025

Technologies: React Native, Expo, OpenWeatherMap API

Built a cross-platform Weather App using React Native, Expo Go and OpenWeatherMap API to deliver real-time weather updates with responsive UI.

INNOHUB

August 2024

Technologies: Flutter, Dart, Firebase

Developed INNOHUB using Flutter, Dart and Firebase to support academic project collaboration with skill-matching, real-time chat, task management and secure file sharing.

SURPLUS SHARE

June 2024

Technologies: HTML, CSS, JavaScript, Python, Django, WebSocket, QR Code Generator

Developed SURPLUS SHARE, a food donation platform using HTML, CSS, JavaScript, Django and WebSocket with real-time chat and QR code integration to connect donors and NGOs.

WORKSHOP

Masters in Embedded Systems Pumo	2025
AI tools and ChatGPT Be10x	2025
Data Science Adverk Technologies	2023

CERTIFICATIONS

Introduction to Industry 4.0 and Industrial Internet of Things NPTEL – ELITE + SILVER (IIT Kharagpur)	2026
Foundations of Deep Learning: Concepts and Applications NPTEL - ELITE (IISc Bangalore)	2026
AI Upskilling Certificate: Technical Foundation Qualcomm Academy	2026
Generative AI for Educators Google	2025
Cloud Practitioner Essentials AWS	2025
Foundation of Cloud IoT Edge ML NPTEL – ELITE (IIT Patna)	2025
Introduction to Front End Development – Simplilearn SkillUP	2025
Digital Circuits NPTEL – IIT Kharagpur	2024
Sensor Technologies: Physics, Fabrication, and Circuits NPTEL – ELITE + SILVER (IISERB)	2024
IOT and Robotics Adverk Technologies	2024
Learn Python Programming – Beginner to Master Udemy	2024
Mastering Data Structure and Algorithms Using C and C++ Udemy	2024
Introduction to Programming in C NPTEL - ELITE (IIT Kanpur)	2024
C, C++, Python with Data Science and Java – COGTIS Technologies Pvt Ltd Code Galatta	2023
Fundamentals of Digital Marketing Google	2023

TECHNICAL SKILLS

Programming & Scripting	C C++ Python JAVA Node.js React.js
Core	Embedded Systems and IOT Data Structures and Algorithms OOPs
Web Development	MERN Stack HTML CSS Javascript
App Development	Flutter
Framework	Express.js
Microcontrollers	STM32 Raspberry Pi ESP32 PIC16F877A ESP8266 Arduino UNO
Databases	MySQL MongoDB
IDE	Keil uVision5 Proteus 8 Professional MP Lab NI Multisim Modelsim Code Blocks Google-Colab VScode Eclipse
Communication Protocols	GSM UART SPI I2C
Tools	RFID WIX MIT App Inventor

ACHIEVEMENTS

SIH 2025	Finalist Dec 2025
Paper Publication	INNOHUB April 2025
Dhruva 2025	2nd Prize Karpagam College of Engineering Project Expo March 2025

Freshathon	2nd Prize Sri Eshwar College of Engineering Project Expo June 2024
Code Fiesta	4th Prize Sri Eshwar College of Engineering C Buildathon Jan 2024
Creatathon	1st Prize Sri Eshwar College of Engineering Portfolio Fest Jan 2024
Quiz	2nd Prize Freshwarite Sri Eshwar College of Engineering 2023
Hacker rank	5 Star for C++ 4 Star for C 3 Star for Python 2 Star for Problem Solving
Skill rack	Programs Solved – 700 Code Tutorials – 505 Code Tracks – 183 Certificates – 3 Bronze – 187 Rank – 41973
Leetcode	Rank - 174888 Contest Rating - 1471 Badges – 3 Problem Solved – 505 Top – 53.93%
Student Mentor	Sri Eshwar College of Engineering June 2024 – Dec 2025